AP CALCULUS AB

Unit 1 • Lesson 1.13 practice

Lesson 1.13 • Removing Discontinuities

4 multiple-choice questions and 1 free-response question

Question 1

Let $g(x) = \dfrac{x^2 - 16}{x - 4}$. What value of $g(4)$ makes g continuous at $x = 4$?

(A) 0
(B) 4
(C) 16
(D) 8

Question 2

For what value of k is $f(x) = \begin{cases} kx^2, & x \le 1 \\ x + 3, & x > 1 \end{cases}$ continuous?

(A) 4
(B) 3
(C) 1
(D) 2

Question 3

For what value of a is $f(x) = \begin{cases} ax + 2, & x < 3 \\ x^2 - a, & x \ge 3 \end{cases}$ continuous?

(A) $\dfrac{11}{4}$
(B) $\dfrac{7}{4}$
(C) $\dfrac{7}{2}$
(D) 7

Question 4

Let $f(x) = \dfrac{\sin x}{x}$ for $x \neq 0$. What value of $f(0)$ makes f continuous at 0?

(A) 0
(B) undefined
(C) 1
(D) π

Question 5 (Free response)

Let $f(x) = \begin{cases} 2x + a, & x < 1 \\ bx^2 + 1, & x \ge 1 \end{cases}$ where a and b are constants, and $f(2) = 9$.

(a) Find b.
(b) Find the value of a that makes f continuous at $x = 1$.
(c) With these values, verify that f is continuous at $x = 1$ using the definition.